

CYBERSECURITY LEARNING PORTFOLIO

SIEM and Analyst Relationship

Introduction to SOC - LetsDefend

Alert monitoring • Investigation • Classification • Improvement



SOC Fundamentals Course Learning Portfolio

A personal record of what I learned from the SIEM lesson and the remaining SOC Fundamentals modules, including Log Management, EDR, SOAR, Threat Intelligence, analyst mistakes, and the final quiz.

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Platform	LetsDefend
Course	SOC Fundamentals (complete course portfolio)
Completion date	July 2026

Portfolio note

I wrote this portfolio after working through the lesson. The aim is not simply to repeat definitions, but to show that I understand how a SIEM alert moves through a SOC and why an analyst's judgement matters.

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1. Course Profile

Item	Details
Course / learning path	SOC Fundamentals
Lesson	SIEM and Analyst Relationship
Platform	LetsDefend
Topic area	Security Operations, SIEM, alert triage, and investigation workflow
Learning method	Lesson reading, simulated SOC interface, practical review, and assessment question

Why I took this lesson

I took this lesson because SIEM is one of the main tools used in a Security Operations Center, and I wanted to understand what an analyst actually does after an alert appears. Before the lesson, I tended to think of an alert as proof that something malicious had happened. The lesson made the process clearer an alert is only a signal. The analyst still has to check the evidence, understand the context, and decide whether the activity is genuinely dangerous.

What the lesson focuses on

The lesson explains what a SIEM is, how rules and filters turn raw events into alerts, and how those alerts are handled in the LetsDefend Monitoring page. It also introduces the shared Main Channel, the Investigation Channel, the alert details view, the Take Ownership action, and the Closed Alerts area. The final example shows why false positives happen and why analysts should report weak detection logic instead of repeatedly closing the same noisy alert.

Learning goals in my own words

- Understand what a SIEM collects and why security teams centralize logs and events.
- Follow an alert from the shared queue to investigation and final closure.
- Know what information to collect before deciding whether an alert is real or false.
- Understand how alert ownership supports teamwork inside a SOC.
- Recognize weak or overly broad detection rules and explain how they can be improved.

2. My Understanding of SIEM

SIEM stands for Security Information and Event Management. In practical terms, it is a central place where security-related events from many systems can be collected, searched, compared, and turned into alerts. Instead of asking an analyst to check every workstation, server, firewall, and application separately, the SIEM brings important information together.

The main purpose is not simply to store logs. A useful SIEM helps the security team notice patterns that may be difficult to see in a single event. For example, one failed password attempt is normal. A burst of failed attempts against the same account, followed by a successful login from an unfamiliar address, deserves attention.

What a SIEM normally does

Capability	What it means to me as an analyst
Log and event collection	It brings events from endpoints, servers, network devices, identity systems, applications, and cloud services into one place.
Normalization	It converts different log formats into a structure that is easier to search and compare.
Search and filtering	It helps me narrow a large amount of data to the user, host, IP address, time range, or indicator I am investigating.
Correlation	It links separate events that may form one suspicious sequence.
Detection rules and thresholds	It defines the conditions that should create an alert.
Dashboards and alerts	It presents activity in a way that helps the SOC decide what needs attention first.

My key takeaway

A SIEM can tell me that something matched a rule, but it cannot understand every business situation on its own. Human review is still necessary.

3. How SIEM Alerts Are Created

An alert is normally produced when incoming events match a rule, filter, correlation, or threshold. The rule represents behaviour that the security team wants an analyst to review. It may be based on a single high-risk event, or it may combine several smaller events that become suspicious when they happen together.

Example from the lesson: repeated password attempts

A person can mistype a password once or twice. That alone is not strong evidence of an attack. However, twenty incorrect passwords within ten seconds are unlikely to be normal manual behaviour. A SIEM rule can count the attempts, compare them with a threshold, and create an alert when the number is exceeded.

1. An authentication system records each login attempt.
2. The events are sent to the SIEM.
3. A rule counts failed attempts for the same account, device, or source.
4. The count crosses the defined threshold.
5. The SIEM creates an alert and places it in the analyst's queue.
6. The analyst checks whether it was user error, an automated process, a misconfiguration, or a genuine attack.

Why an alert is not the final answer

Rules work with the information available to them. They may not know that an administrator is carrying out approved testing, that a business application retries automatically, or that a user is travelling. This is why the analyst must add context. A technically correct rule match can still be a false positive.

Examples of SIEM products mentioned in the lesson

The lesson refers to IBM QRadar, ArcSight ESM, FortiSIEM, and Splunk. The menus and terminology may differ from one platform to another, but the analyst's core responsibility remains similar: review the alert, collect supporting evidence, record a clear conclusion, and escalate when necessary.

4. The Relationship Between SIEM and the SOC Analyst

In many organisations, security engineers or detection engineers are responsible for connecting data sources, maintaining the SIEM, and writing correlation rules. The SOC analyst usually works with the alerts produced by that engineering work. The analyst's job begins where automated detection stops.

When I open an alert, I need to move beyond the title. I should understand what triggered it, which system or user is involved, whether the activity succeeded, and whether other tools support the same conclusion. This is the difference between simply acknowledging alerts and performing a proper investigation.

Questions I should ask during triage

- What exact event or behaviour triggered the rule?
- Which user, host, IP address, process, file, URL, or application is involved?
- When did it happen, and are the timestamps consistent across systems?
- Was the action successful, blocked, or only attempted?
- Is the behaviour normal for this user or device?

- Do endpoint, network, identity, or application logs support the alert?
- Are there other affected systems or related events?
- Does the case need closure, monitoring, escalation, or immediate containment?

Analyst mindset

The alert title gives me a direction, not a verdict. I should be able to explain what evidence I checked and why that evidence supports my conclusion.

5. Guide to Every Dashboard and Interface Area in the Lesson

The lesson mainly uses the LetsDefend Monitoring page to demonstrate how alerts are organised. The live platform may change its layout over time, but the workflow described in the lesson remains useful. Below is my explanation of every page, channel, and panel that is shown or referenced.

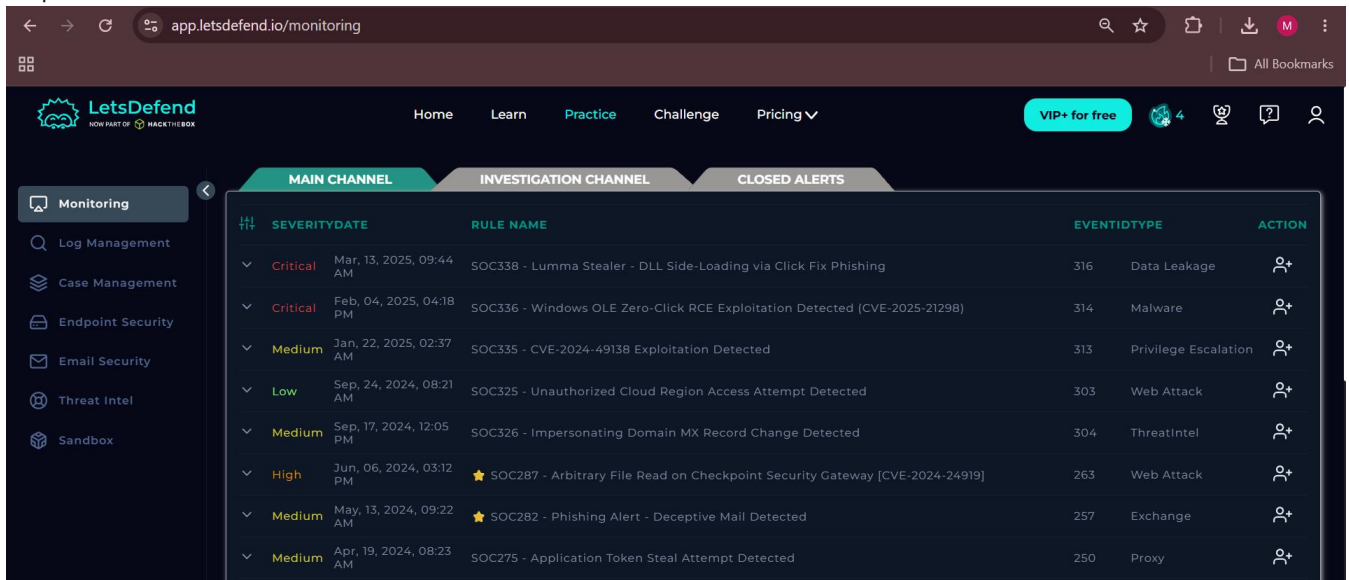
Area	Main purpose
Monitoring page / SIEM dashboard	The central workspace where alerts can be reviewed and moved through the investigation lifecycle.
Main Channel	The shared queue for newly created alerts that have not yet been taken by an analyst.
Action area / Take Ownership	The control used to assign an alert to the analyst who will investigate it.
Investigation Channel	The working queue for alerts that are currently being analyzed.
Alert Details view	The panel or page containing the technical information needed to start the investigation.
Closed Alerts	The channel where completed and closed alerts can be accessed later.
Questions and Progress area	The lesson assessment area showing the question, answer field, and completion progress.

5.1 Monitoring Page / SIEM Dashboard

The Monitoring page is the main SIEM-style dashboard in the LetsDefend simulation. It brings alerts and their current status into one workspace. From here, an analyst can see which alerts are new, which ones are already being investigated, and which ones have been closed.

In a real SOC, this page is important because analysts work as a team. A shared view reduces confusion, helps the team prioritize urgent cases, and makes it easier to see the overall workload. I would use the Monitoring page as the starting point for every shift: review the queue, check severity and age, and select a case that matches my

responsibilities.



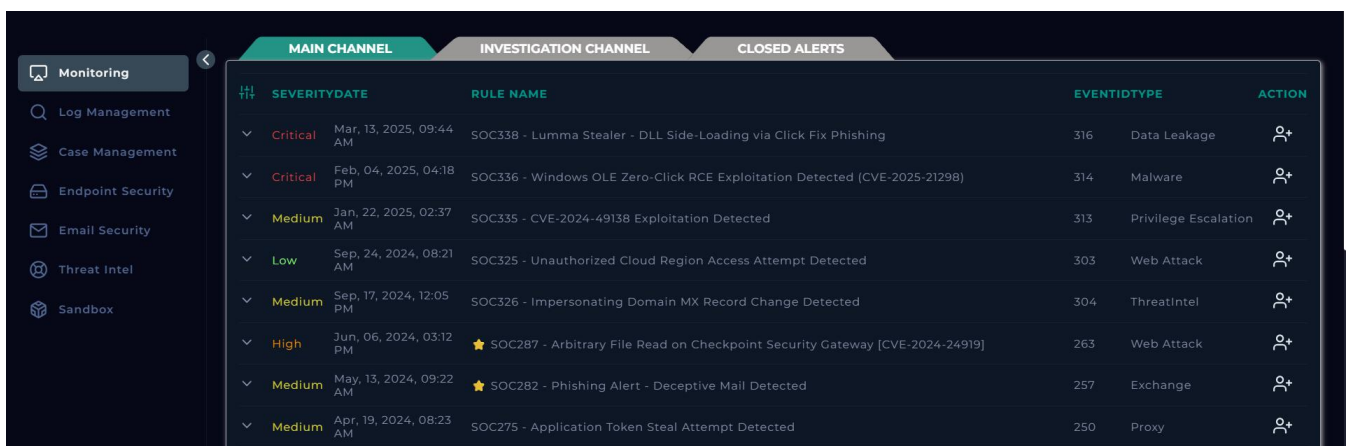
What to capture in a screenshot

Show the full Monitoring page with the channel tabs visible. Avoid exposing your email address, account details, or any sensitive information.

5.2 Main Channel

The Main Channel is the shared queue for newly generated alerts. At this stage, the alert is available for an analyst to review and claim. The lesson compares it with a shared team channel because everyone can see the same pool of unassigned work.

The most important point is coordination. If two analysts investigate the same alert without knowing about each other, time is wasted. The Main Channel makes the work visible. Before selecting an alert, I would normally look at the alert name, severity, timestamp, affected system, and whether a more urgent case is already waiting.

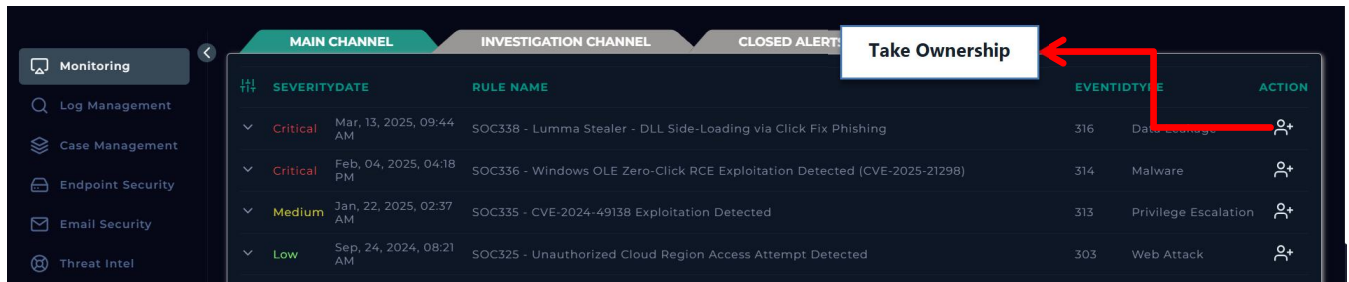


5.3 Action Area and Take Ownership

The Action area contains the control used to take ownership of an alert. When I click Take Ownership, I am accepting responsibility for that case. The alert is then moved away from the shared unassigned queue and into the Investigation Channel.

This simple action has an operational purpose: it shows my teammates that the alert is being handled. It also creates accountability because the case is now associated with a particular analyst. In a professional SOC, I should

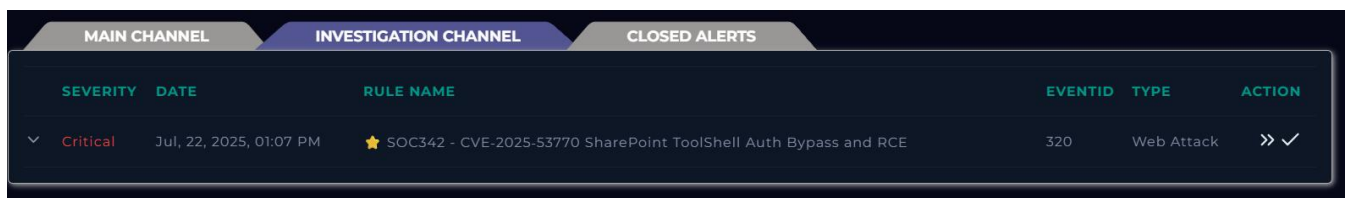
not take ownership and leave the alert untouched; once assigned, I should begin triage and keep the case notes current.



5.4 Investigation Channel

The Investigation Channel contains alerts that an analyst has already claimed and is actively working on. It represents work in progress rather than new work. This is where I would return to continue an investigation, add findings, or review a case before closing or escalating it.

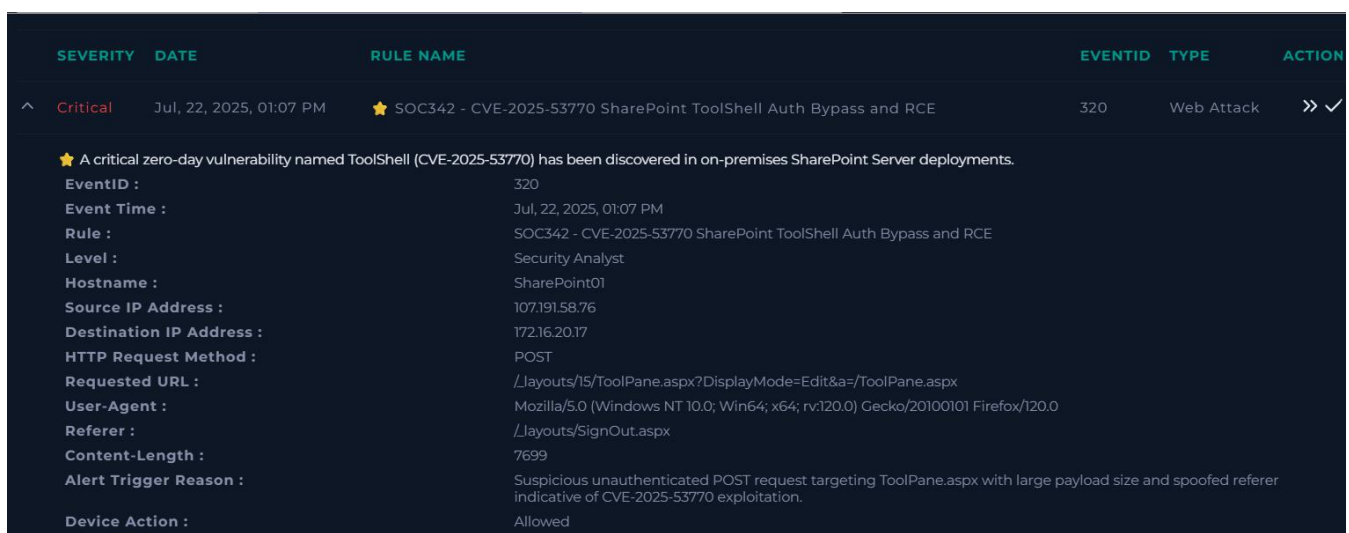
The channel also helps the team understand capacity. If several alerts are already in investigation, the SOC manager or shift lead can see that analysts may be busy. A well-managed investigation queue should not become a place where alerts remain without updates.



5.5 Alert Details View

When an alert is opened, the details view provides the initial evidence needed to understand the case. The exact fields depend on the alert, but the lesson mentions information such as the host name, IP address, and file hash. Other useful fields can include the username, URL or domain, event time, alert severity, detection rule, and a short description of the activity.

I would use these fields to create the first scope of the investigation. For example, the host name tells me which device to check in EDR, the IP address gives me an indicator to search in network logs or threat intelligence, and the timestamp helps me build a timeline. The details view is the starting evidence, not the complete investigation.



5.6 Closed Alerts Channel

The Closed Alerts channel stores alerts after the analyst has finished the case and closed it. This is the answer to the lesson question. Closed alerts remain useful because they provide a record of what was investigated, what decision was made, and why the case was closed.

In a real environment, I may return to a closed alert when a similar event happens again, when a senior analyst reviews my work, or when an audit requires evidence of how an alert was handled. For that reason, closing notes should be clear enough for someone else to understand the decision later.

Close Alert

Event ID :
320

☒ True Positive ☐ False Positive

Note:
Suspicious unauthenticated POST request.

Close Alert

MAIN CHANNEL			INVESTIGATION CHANNEL			CLOSED ALERTS		
SEVERITY	DATE CLOSED	RULE NAME	EVENTID	TYPE	RESULT	ACTION		
▼ Critical	Jul, 16, 2026, 02:32 PM	★ SOC342 - CVE-2025-53770 SharePoint ToolShell Auth Bypass and RCE	320	Web Attack	✓	↺		

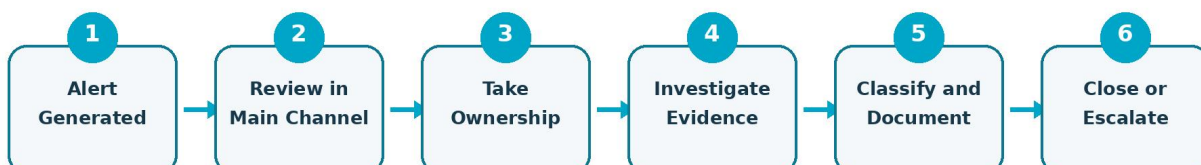
5.7 Questions and Progress Area

At the end of the lesson, the Questions area checks whether I understood the dashboard workflow. It shows the current progress, the question, the required answer format, and the submission control. This area is not part of the operational SOC dashboard, but it is part of the training interface and provides evidence that the lesson was completed.

Important distinction

Main Channel means unassigned work, Investigation Channel means active work, and Closed Alerts means completed work. Together, these areas show the life cycle of an alert.

6. Alert Investigation Workflow



Key principle: the alert is the starting point - evidence and context determine the conclusion.

Figure 1: The basic alert life cycle from new alert to investigation and closure.

Review the Main Channel: I start by checking new alerts and considering their severity, age, affected asset, and potential business impact.

Take ownership: I assign the alert to myself so that the team knows who is responsible for the case.

Open the alert details: I collect the initial identifiers, including the host, user, IP addresses, file information, URL, and timestamp.

Gather supporting evidence: I use EDR, log management, network data, identity records, threat intelligence, or other relevant tools.

Build a timeline and scope: I determine what happened first, whether the activity succeeded, and whether more than one system is affected.

Classify the alert: I decide whether it is malicious, benign, a policy issue, a false positive, or a case that needs more investigation.

Document the reasoning: I record what I checked, what I found, and why I reached the conclusion.

Close or escalate: I close completed alerts or escalate serious, widespread, or technically complex incidents to the appropriate person.

What good case notes should contain

- A short summary of the alert and the reason it triggered.
- The affected user, device, account, or application.
- The indicators and logs that were checked.
- A timeline of the most important events.
- The final classification and supporting evidence.
- Any containment, escalation, or follow-up action.
- A clear reason for closure if the activity was benign or false.

7. True Positives, False Positives, and Rule Feedback

True positive

A true positive means the alert correctly identified malicious or unauthorised activity. For example, a repeated-login alert may be confirmed as password guessing from an external address, followed by a successful login and suspicious account activity.

False positive

A false positive means the rule matched legitimate activity. The lesson uses a useful example: a rule searches URLs for the word "union" because UNION can appear in SQL injection attacks. A user searches Google for information about SQL UNION usage, so the word appears in the URL and the SIEM creates an alert. The rule matched the keyword, but the activity itself was harmless.

What I learned from the UNION example

The example shows that detection logic needs context. A single word is not enough to prove an attack. A better rule might check whether the word appears with other SQL syntax, whether the request targets a business application rather than a search engine, whether unusual response codes appear, or whether the same source performs several suspicious requests.

Why analyst feedback matters

If I repeatedly see the same harmless alert, simply closing it every time is not a good long-term solution. I should explain the pattern to the SIEM or detection team. They can then adjust the rule, add a safe exclusion, change the threshold, or include more context. This reduces alert fatigue and leaves analysts more time for genuine threats.

Weak habit	Better approach
Closing an alert because the title looks familiar	Open it, confirm the evidence, and compare it with previous cases.
Treating every rule match as an attack	Check the user, host, timing, business context, and supporting logs.
Ignoring repeated false positives	Report the pattern and recommend rule tuning.
Writing "false positive" with no explanation	Record the exact evidence that shows the activity was legitimate.
Depending on one tool only	Correlate evidence across the SIEM, endpoint, network, identity, and threat intelligence sources.

8. Course Question and Answer

Question from the lesson

When you close an alert, which channel (tab on the Monitoring page) can you access it from?

Correct answer

Correct answer	Closed Alerts
Answer format	***** *****

Correct

When you close an alert, which channel(tab on the monitoring page) can you access it from?

closed alerts

Completed

Get unstuck?

Why this answer is correct

Once an analyst has completed the investigation and closes the case, the alert is no longer active in the Main Channel or the Investigation Channel. It can be accessed from the Closed Alerts channel. This keeps the active queues focused on current work while preserving completed cases for review, quality checks, future comparison, and audit history.

How I would answer it in the platform

1. Open the Monitoring page.
2. Look at the available channel or tab names.
3. Identify the tab used for completed cases.
4. Enter "Closed Alerts" in the answer box.
5. Submit the answer and capture the result as portfolio evidence.

What this question tests

The question is simple, but it checks whether I understand the difference between new, active, and completed alerts in the SOC workflow.

9. What I Learned and the Skills I Developed

Knowledge I gained

After completing this lesson, I understand that a SIEM is both a data platform and an operational starting point for investigations. I can explain how events become alerts, how analysts claim and investigate those alerts, and why a final decision must be based on evidence rather than the rule name alone.

Practical skills

- Navigating the Monitoring page and recognizing the purpose of each channel.
- Taking ownership of an alert and understanding why assignment matters.
- Reading initial alert fields such as host name, IP address, file hash, URL, severity, and time.
- Planning which supporting tools and logs should be checked next.
- Separating a true threat from normal or harmless activity.
- Writing a defensible closure reason and knowing when to escalate.
- Recognizing detection-rule problems and providing useful feedback.

Skills matrix

Skill area	Evidence from this lesson	Current level
SIEM fundamentals	I can explain collection, correlation, thresholds, and alerts.	Foundational
Alert triage	I can follow the alert from Main Channel to investigation.	Foundational
Analytical thinking	I can explain why context is needed before classifying an alert.	Developing
SOC workflow	I understand ownership, active investigation, closure, and escalation.	Foundational
Documentation	I know what information should be recorded in case notes.	Developing
Detection improvement	I can describe how false-positive feedback supports rule tuning.	Foundational

10. Personal Reflection and Real-World Application

Personal reflection

The most useful part of this lesson was learning not to trust an alert blindly. A SIEM is powerful because it can process more data than a person could review manually, but a rule can still be too broad or miss important context. I now see the analyst as the person who turns a technical signal into a justified security decision.

I also understood the practical reason for alert ownership. It may look like a small interface action, but it prevents duplicated work and makes responsibility visible. In a busy SOC, disciplined workflow is just as important as technical knowledge.

The false-positive example was memorable because it showed how easily a keyword rule can create noise. It reminded me that good investigation requires curiosity: I should ask why the rule triggered, what actually happened, and what evidence would prove or disprove the suspected attack.

How I could use this knowledge at work

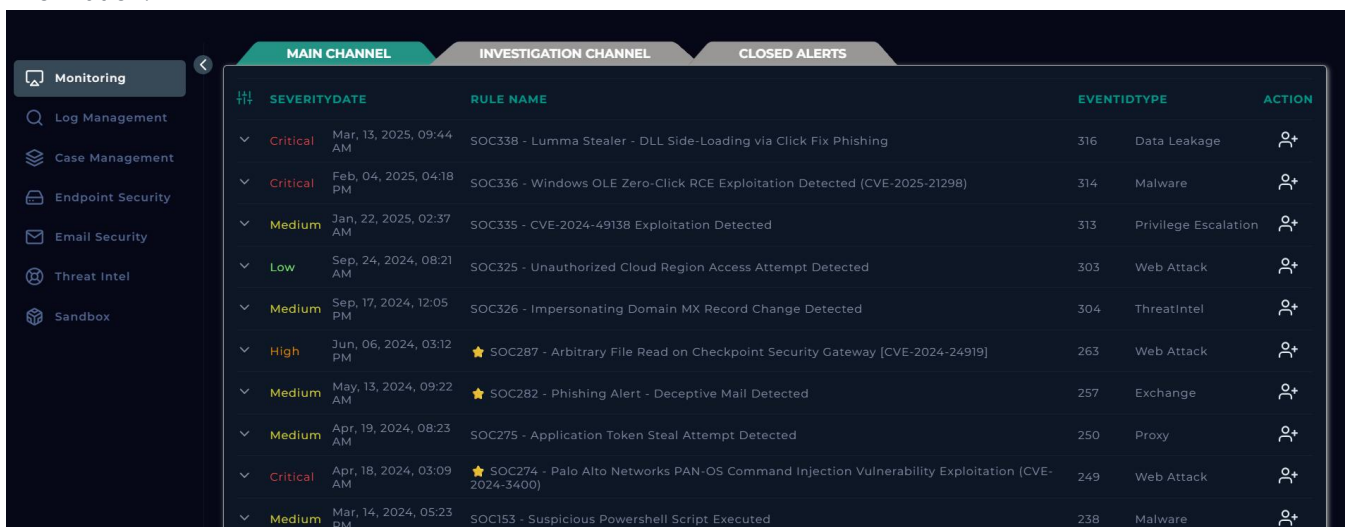
- Review suspicious login, malware, URL, file, or network alerts during an entry-level SOC shift.
- Use alert details to decide which endpoint, identity, network, or threat-intelligence data to check.
- Document a clear timeline and final classification for another analyst or incident responder.
- Escalate cases involving privileged accounts, multiple systems, confirmed malware, or possible data loss.
- Identify repeated false positives and explain the evidence to the detection engineering team.

What I still need to practise

This lesson gives me the workflow, but I still need hands-on practice with real log searches, endpoint investigation, network analysis, and case documentation. My next goal is to investigate complete SIEM scenarios where I can collect evidence from several tools and defend my conclusion.

11. Screenshot Evidence

The following placeholders are ready for screenshots from my own Lets Defend account. I will replace each placeholder with a clear image and keep the caption underneath. Before adding screenshots, I will crop unnecessary browser areas and hide email addresses, account details, tokens, or any sensitive information.



SEVERITY	DATE	RULE NAME	EVENTID	TYPE	ACTION
Critical	Mar, 13, 2025, 09:44 AM	SOC338 - Lumma Stealer - DLL Side-Loading via Click Fix Phishing	316	Data Leakage	+
Critical	Feb, 04, 2025, 04:18 PM	SOC336 - Windows OLE Zero-Click RCE Exploitation Detected (CVE-2025-21298)	314	Malware	+
Medium	Jan, 22, 2025, 02:37 AM	SOC335 - CVE-2024-49138 Exploitation Detected	313	Privilege Escalation	+
Low	Sep, 24, 2024, 08:21 AM	SOC325 - Unauthorized Cloud Region Access Attempt Detected	303	Web Attack	+
Medium	Sep, 17, 2024, 12:05 PM	SOC326 - Impersonating Domain MX Record Change Detected	304	ThreatIntel	+
High	Jun, 06, 2024, 03:12 PM	★ SOC287 - Arbitrary File Read on Checkpoint Security Gateway (CVE-2024-24919)	263	Web Attack	+
Medium	May, 13, 2024, 09:22 AM	★ SOC282 - Phishing Alert - Deceptive Mail Detected	257	Exchange	+
Medium	Apr, 19, 2024, 08:23 AM	SOC275 - Application Token Steal Attempt Detected	250	Proxy	+
Critical	Apr, 18, 2024, 03:09 AM	★ SOC274 - Palo Alto Networks PAN-OS Command Injection Vulnerability Exploitation (CVE-2024-3400)	249	Web Attack	+
Medium	Mar, 14, 2024, 05:23 PM	SOC153 - Suspicious Powershell Script Executed	238	Malware	+

Monitoring dashboard overview. This image should show the main alert workspace and the available channel tabs.

MAIN CHANNEL		INVESTIGATION CHANNEL		CLOSED ALERTS	
SEVERITY	DATE	RULE NAME	EVENTID	TYPE	ACTION
▼ Critical	Mar, 13, 2025, 09:44 AM	SOC338 - Lumma Stealer - DLL Side-Loading via Click Fix Phishing	316	Data Leakage	👤+
▼ Critical	Feb, 04, 2025, 04:18 PM	SOC336 - Windows OLE Zero-Click RCE Exploitation Detected (CVE-2025-21298)	314	Malware	👤+
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▼ Low	Sep, 24, 2024, 08:21 AM	SOC325 - Unauthorized Cloud Region Access Attempt Detected	303	Web Attack	👤+
▼ Medium	Sep, 17, 2024, 12:05 PM	SOC326 - Impersonating Domain MX Record Change Detected	304	ThreatIntel	👤+
▼ High	Jun, 06, 2024, 03:12 PM	★ SOC287 - Arbitrary File Read on Checkpoint Security Gateway [CVE-2024-24919]	263	Web Attack	👤+
▼ Medium	May, 13, 2024, 09:22 AM	★ SOC282 - Phishing Alert - Deceptive Mail Detected	257	Exchange	👤+
▼ Medium	Apr, 19, 2024, 08:23 AM	SOC275 - Application Token Steal Attempt Detected	250	Proxy	👤+
▼ Critical	Apr, 18, 2024, 03:09 AM	★ SOC274 - Palo Alto Networks PAN-OS Command Injection Vulnerability Exploitation (CVE-2024-3400)	249	Web Attack	👤+

Main Channel This image should show a newly generated alert before ownership is taken.

MAIN CHANNEL		INVESTIGATION CHANNEL		CLOSED ALERTS	
SEVERITY	DATE	RULE NAME	EVENTID	TYPE	ACTION

Investigation Channel. This image should show an alert that has been assigned and is actively being analyzed.

MAIN CHANNEL		INVESTIGATION CHANNEL		CLOSED ALERTS	
SEVERITY	DATE CLOSED	RULE NAME	EVENTID	TYPE	RESULT ACTION
▼ Critical	Jul, 16, 2026, 02:32 PM	★ SOC342 - CVE-2025-53770 SharePoint ToolShell Auth Bypass and RCE	320	Web Attack	✓ ↺

Closed Alerts. This image should show the channel where completed alerts can be accessed.

12. SIEM Glossary and Sources

Glossary

Term	Meaning
SOC	Security Operations Center: the people, processes, and technology responsible for monitoring and responding to security threats.
SIEM	Security Information and Event Management: a platform that collects events, applies detection logic, and supports investigation.
Alert	A notification created because activity matched a detection rule, threshold, or correlation.
Main Channel	The shared queue for new, unassigned alerts.
Investigation Channel	The queue for alerts that have been claimed and are actively being investigated.

Term	Meaning
Closed Alerts	The channel containing completed alerts after closure.
True positive	An alert that correctly identifies malicious or unauthorised activity.
False positive	An alert that identifies legitimate activity as suspicious.
Triage	The first review used to understand, prioritise, and classify an alert.
Escalation	Passing a case to a senior analyst, incident responder, or specialist because of its risk or complexity.

Source and access note

The live LetsDefend lesson page requires an authenticated account. This portfolio was therefore written primarily from the lesson text and notes supplied for this project. The wording has been rewritten into a personal learning record rather than copied as course text.

Lesson page: [LetsDefend - SIEM and Analyst Relationship](#)

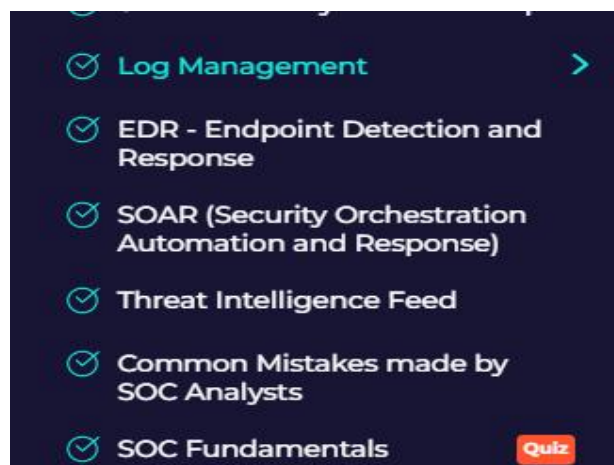
Answer confirmation: [LetsDefend Community FAQ - SOC Fundamentals](#)

Recommended next steps

- Complete a full SIEM alert investigation scenario.
- Practise Windows Event Log and Linux log analysis.
- Learn the basics of EDR and endpoint process investigation.
- Use threat-intelligence sources to check IP addresses, domains, URLs, and file hashes.
- Study incident severity, escalation criteria, and case documentation.
- Map common alerts to MITRE ATT&CK techniques.

13. Complete SOC Fundamentals Course Map

The earlier pages focus on the SIEM and analyst relationship because that lesson explains the basic alert lifecycle. The complete SOC Fundamentals course goes further. It shows how an analyst uses logs, endpoint data, automation, and threat intelligence to turn a SIEM alert into a reasoned decision. The image below is the course list I used to organise this expanded portfolio.



Remaining lessons shown in the LetsDefend SOC Fundamentals course.

Lesson	How it supports an investigation
Log Management	Provides central access to web, operating system, firewall, proxy, DNS, and endpoint logs so I can search activity across the environment.
EDR - Endpoint Detection and Response	Shows what happened on a workstation or server, including processes, command lines, files, network connections, and response actions.
SOAR	Connects security tools, automates repeatable checks, and uses playbooks to make investigations faster and more consistent.
Threat Intelligence Feed	Adds external context about hashes, IP addresses, domains, URLs, malware families, and command-and-control infrastructure.
Common Mistakes	Explains habits that can produce weak conclusions, such as trusting one tool, rushing sandbox analysis, or ignoring related logs.
SOC Fundamentals Quiz	Checks whether I can connect the SOC structure, analyst role, tools, workflow, and incident-response concepts.

How the tools fit together

SIEM tells me what matched a rule. Log Management helps me search the wider environment. EDR explains endpoint behaviour. Threat Intelligence adds reputation and external context. SOAR brings the steps together. My responsibility is to compare all of that evidence before I close or escalate the case.

14. Log Management

What I learned

Log Management gives the SOC one place to search records produced by many systems. These records may come from web servers, operating systems, firewalls, proxies, DNS services, authentication platforms, applications, and EDR tools. The main benefit is not simply storage. It is the ability to ask one investigation question across several data sources without signing in to every device separately.

For example, after finding a suspicious command-and-control address in a malware investigation, I can search the central logs to see whether any other device contacted the same address. That search can show the source system, destination, port, time, protocol, and raw event. This helps me understand whether I am dealing with one endpoint or a wider incident.

Log Management dashboard explained

Dashboard area	What I use it for
Search and filter controls	I choose a field, an operator such as contains or equals, and a value such as an IP address, URL, port, username, or host name.
Basic and advanced search modes	Basic search is useful for a direct lookup. A more advanced mode is useful when I need several conditions, exclusions, or a narrower time range.
Log results table	This displays matching events. Typical fields include time, source and destination addresses, source and destination ports, log type, action, and device name.
Raw log field	The raw event preserves the original message. I read it when the normalised columns do not contain enough context or when I need to verify exactly what the source system recorded.

Dashboard area	What I use it for
Event details	Opening a result lets me inspect the complete event and collect values for the investigation timeline.
Time filter	I reduce noise by searching around the alert time first, then expand the window if I suspect earlier preparation or later follow-on activity.

How I would investigate with this dashboard

1. Start with one reliable indicator from the alert, such as a URL, IP address, user, hostname, or port.
2. Select the most relevant field. When the exact field is unclear, search the raw log for a distinctive part of the value.
3. Limit the time range around the alert so the first result set is manageable.
4. Open matching events and compare the source, destination, action, protocol, and timestamps.
5. Search related values to find other affected devices or repeated activity.
6. Record the useful events in the case notes and explain what they prove.

My practical takeaway

A single log event rarely tells the whole story. The useful skill is following one value into related events and building a timeline that another analyst can understand.

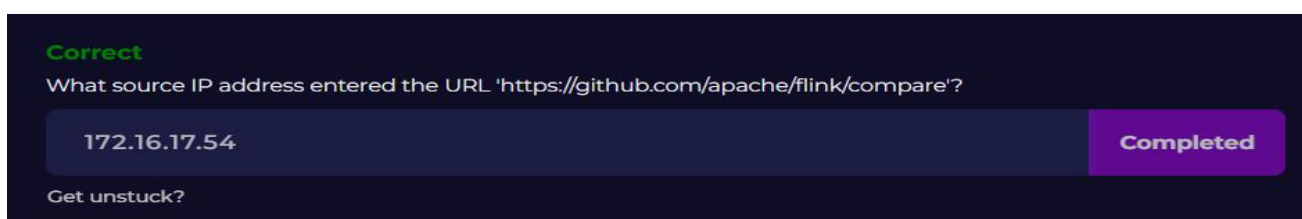
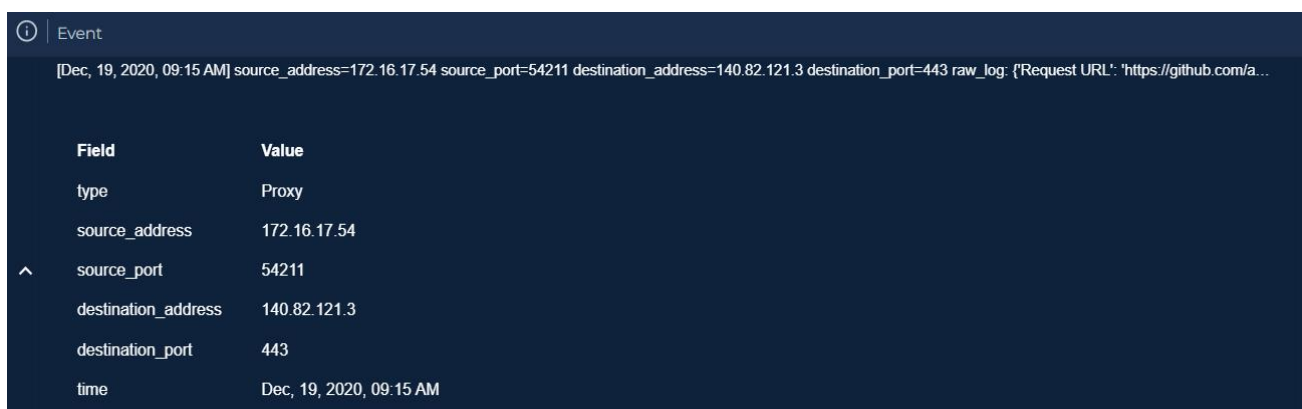
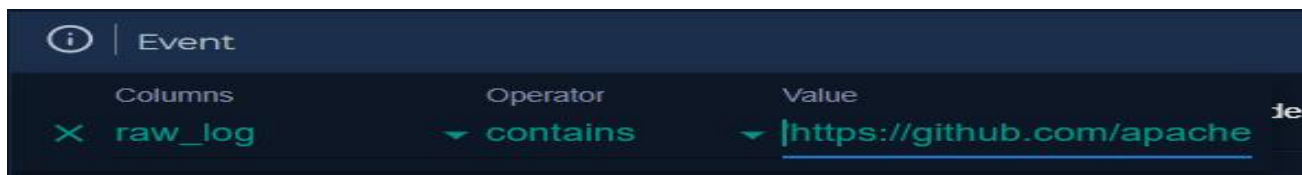
14.1 Log Management Lesson Questions

Question 1

Question: What source IP address entered the URL "https://github.com/apache/flink/compare"?

Answer: 172.16.17.54

Explanation: Open Log Management, filter the raw_log field with the contains operator, and paste the URL or search for a distinctive word such as "compare". Open the matching event and read the source IP field.

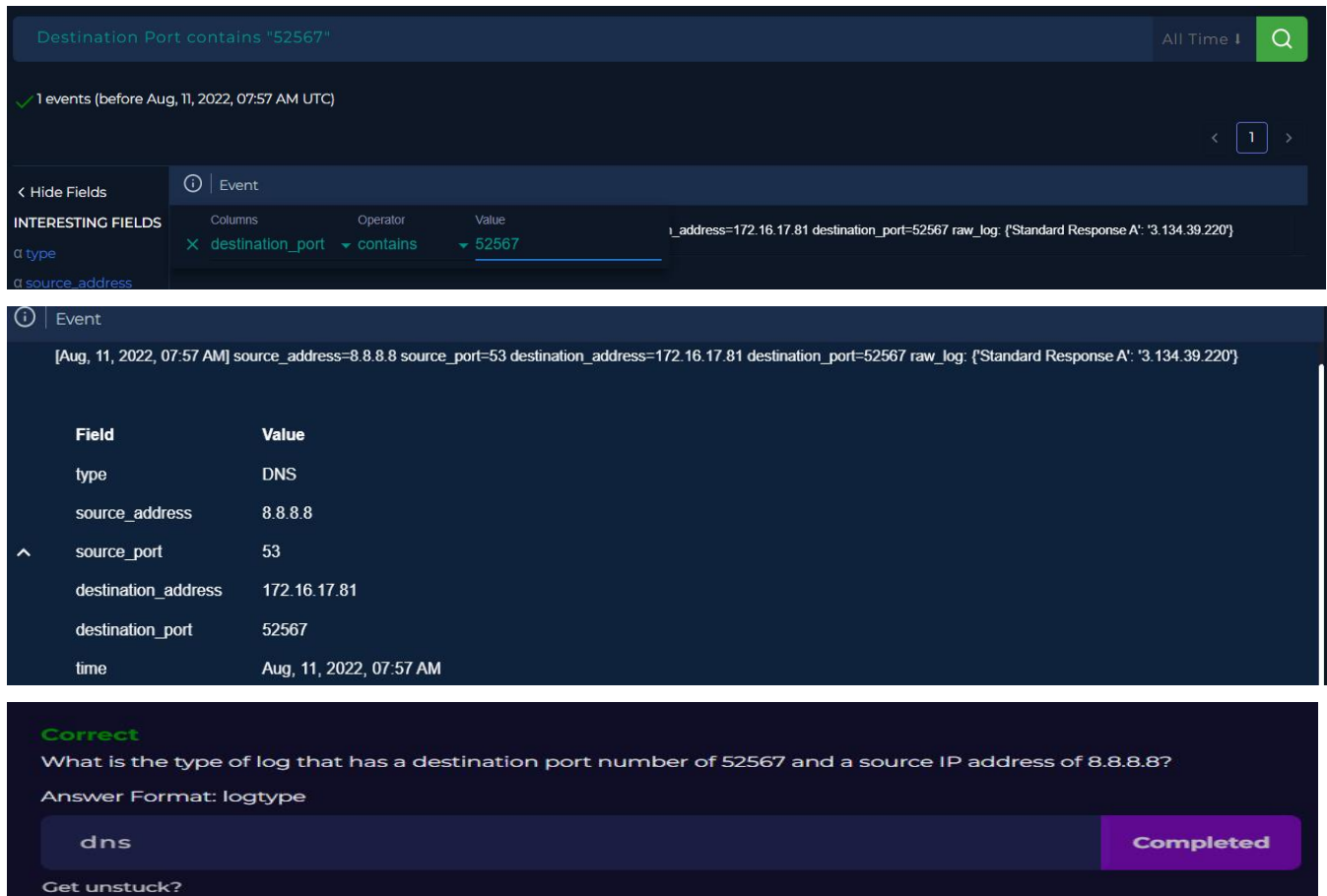


Question 2

Question: What is the log type with destination port 52567 and source IP address 8.8.8.8?

Answer: DNS

Explanation: Search for source IP 8.8.8.8, then locate the event whose destination port is 52567. The event type identifies it as a DNS log.



Destination Port contains "52567" All Time I 

✓ 1 events (before Aug, 11, 2022, 07:57 AM UTC)

< 1 >

< Hide Fields | Event

Columns	Operator	Value
destination_port	contains	52567

u_address=172.16.17.81 destination_port=52567 raw_log: {'Standard Response A': '3.134.39.220'}

INTERESTING FIELDS

type

source_address

Event

[Aug, 11, 2022, 07:57 AM] source_address=8.8.8.8 source_port=53 destination_address=172.16.17.81 destination_port=52567 raw_log: {'Standard Response A': '3.134.39.220'}

Field	Value
type	DNS
source_address	8.8.8.8
source_port	53
destination_address	172.16.17.81
destination_port	52567
time	Aug, 11, 2022, 07:57 AM

Correct

What is the type of log that has a destination port number of 52567 and a source IP address of 8.8.8.8?

Answer Format: logtype

dns

Completed

Get unstuck?

What these questions are testing

The questions are not only testing whether I can copy a value. They test whether I understand field-based searching. The first starts with a URL and asks me to identify the device that made the request. The second combines two conditions and asks me to interpret the type of traffic. In a real SOC, this same approach is used to pivot from an alert into supporting log evidence.

Exercise-data note

The values above match the published SOC Fundamentals exercise dataset. If LetsDefend changes the training data, I would verify the current result in my own dashboard before submitting it.

15. EDR - Endpoint Detection and Response

What I learned

EDR gives an analyst detailed visibility into endpoints such as laptops, desktops, and servers. It continuously records security-relevant activity and helps the SOC investigate what a process did, which command launched it, which files were created, and which network destinations were contacted. This is more useful than relying only on a malware alert because it shows behaviour and context.

The lesson also introduced response capabilities. If an endpoint is confirmed as compromised, an analyst may isolate it from the network so the attacker cannot move laterally or continue communicating with external infrastructure. Containment should still follow the organisation's approval process because isolating a critical server can affect business operations.

Endpoint Security / EDR dashboard explained

Dashboard area	What it tells me
Endpoint search	I search by hostname, IP address, username, file name, or indicator such as a hash to find the relevant device.
Device summary	This identifies the host, operating system, last-seen time, assigned user, and basic security status.
Process list or process tree	I can see which process started another process, the full command line, file path, execution time, and parent-child relationship.
Network connections	This shows which processes connected to external or internal addresses and which ports were used.
Browser history	This can support investigations involving suspicious downloads, phishing links, or malicious websites.
Command or terminal history	This reveals commands executed on the device and can help confirm scripts, tools, or attacker activity.
Live investigation	A permitted analyst can connect to the endpoint to collect additional information or run approved checks.
Containment	The device can be isolated from normal network communication while retaining a controlled security-management connection.
IOC search	A hash, file name, domain, or other indicator can be searched across multiple endpoints to estimate the scope of the incident.

How I read endpoint evidence

- I start with the hostname and time from the SIEM alert, then open the matching endpoint.
- I identify the suspicious process and check its parent process, command line, path, user, and execution time.
- I check whether the process created files, launched child processes, or opened network connections.
- I compare the behaviour with the user's role and normal activity before deciding that it is malicious.
- I search the same indicator across other endpoints to check whether the incident is isolated or widespread.
- I document any containment action, including who approved it and why it was necessary.

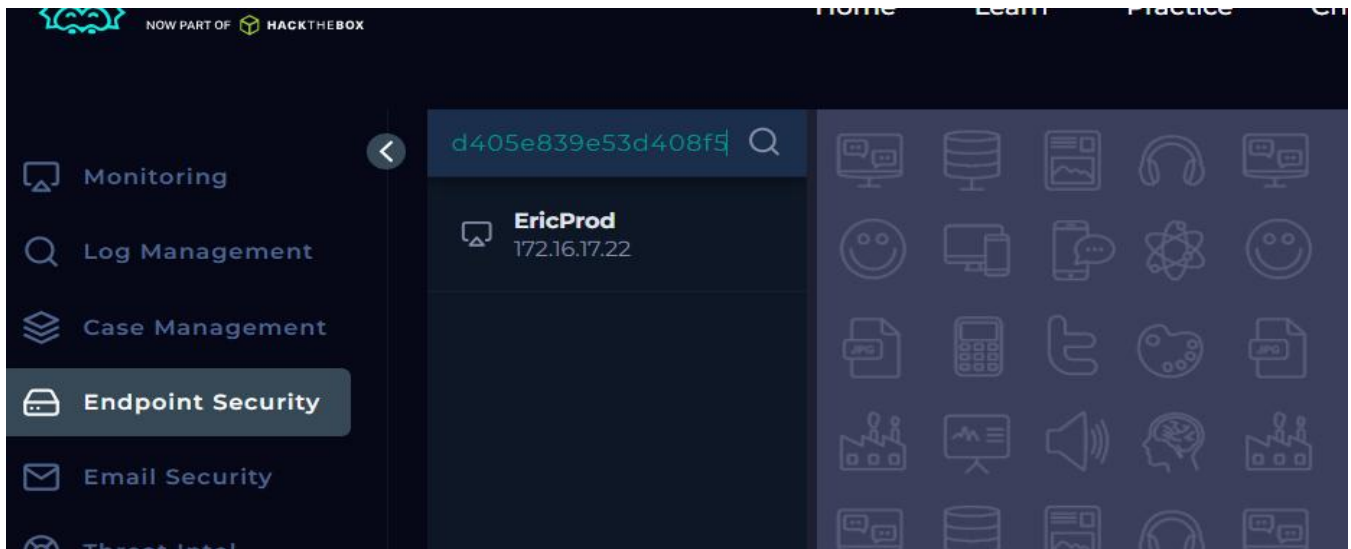
15.1 EDR Lesson Questions

Question 1

Question: What is the host name of the device where the "nmap" file with hash 83e0cfc95de1153d405e839e53d408f5 was executed?

Answer: EricProd

Explanation: Search the hash in the Endpoint Security dashboard, open the matching nmap execution, and read the host name from the endpoint details.



Correct

What is the hostname of the device where the "nmap" file with a hash value of "83e0cfc95de1153d405e839e53d408f5" is executed?

EricProd

Completed

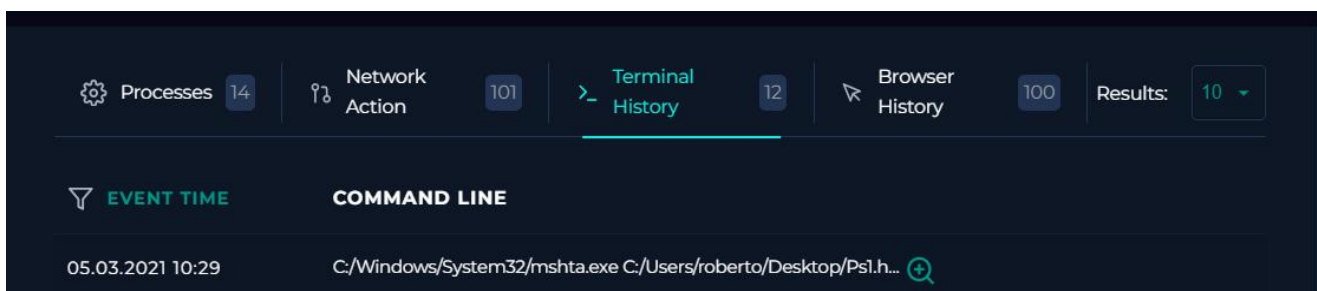
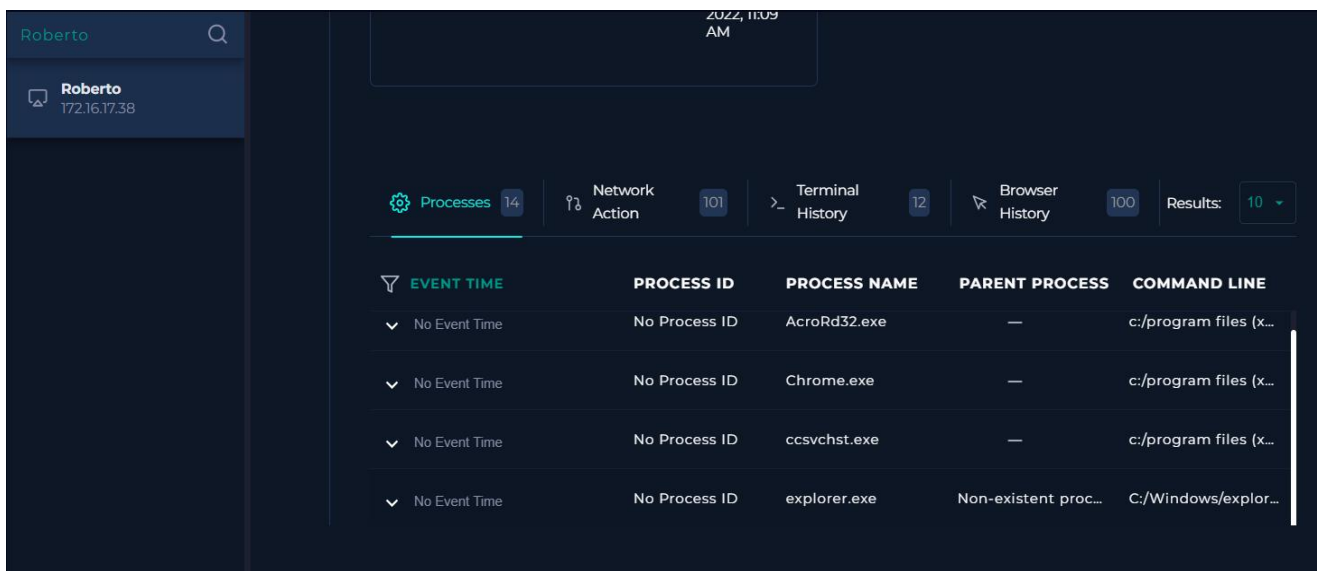
Get unstuck?

Question 2

Question: A "Ps1.hta" file was executed on the device with host name "Roberto". What is the complete CMD command?

Answer: C:/Windows/System32/mshta.exe C:/Users/roberto/Desktop/Ps1.hta

Explanation: Search for the Roberto endpoint, review its processes or command history, and locate the mshta.exe execution that opened Ps1.hta.



Correct

A "Ps1.hta" file was executed on a device with the hostname "Roberto". What is the complete CMD command?

`C:/Windows/System32/mshta.exe C:/Users/roberto/Desktop/Ps1.hta`**Completed**[Get unstuck?](#)

Why the command line matters

A file name by itself can be misleading. The full command line shows which executable launched the file, where the file was stored, and whether extra arguments were used. In this example, mshta.exe is a legitimate Windows binary, but attackers can abuse it to execute HTA content. The command line provides the context needed to decide what happened.

EDR investigation principle

Do not stop at the detection name. Follow the process tree, command line, file activity, network activity, and user context before classifying the endpoint.

16. SOAR - Security Orchestration, Automation and Response

What I learned

SOAR connects security products and turns repeatable investigation steps into organised workflows. Instead of manually copying an IP address into several tools, a SOAR playbook can enrich it with reputation data, create a case, assign tasks, request approval, and record the results. This saves time and makes the investigation process more consistent.

Automation does not remove the analyst from the decision. A good playbook automates predictable tasks and presents the evidence clearly, while important actions such as disabling an account, blocking an address, or isolating a business-critical system may still require human review and approval.

Case Management / SOAR dashboard explained

Dashboard area	How it supports the analyst
Case queue	Shows open cases, ownership, severity, status, age, and priority so the team can see what is being worked on.
Case details	Collects the alert summary, affected assets, indicators, evidence, and final classification in one record.
Playbook tasks	Provides a consistent sequence of checks, such as validating an IP, checking a hash, reviewing endpoint activity, and documenting the decision.
Automated enrichment	Queries connected services for IP reputation, hash reputation, domain information, sandbox results, or other context.
Activity timeline	Records automated actions, analyst notes, status changes, assignments, and response decisions.
Integrations	Connects SIEM, EDR, email security, threat intelligence, ticketing, firewalls, and other products.

Dashboard area	How it supports the analyst
Response actions	Can perform approved actions such as creating a block request, disabling an account, or isolating an endpoint.
Evidence and attachments	Keeps screenshots, logs, files, and supporting notes with the case so the decision can be reviewed later.

A simple phishing playbook example

7. Create a case from the email-security or SIEM alert.
8. Extract the sender, URLs, domains, attachment hashes, recipient, and delivery time.
9. Check the indicators against threat-intelligence sources.
10. Search the mailbox and logs for other recipients or related messages.
11. Check EDR for file execution or suspicious browser activity.
12. Ask the analyst to confirm malicious or benign classification.
13. Run approved containment actions and notify the relevant team.
14. Document the outcome and close the case.

No separate lesson exercise identified

The course material I reviewed explains SOAR in detail but does not show a separate hands-on lesson question. The final course quiz contains the playbook question above.

17. Threat Intelligence Feed

What I learned

A threat intelligence feed provides external information about indicators such as malware hashes, IP addresses, domains, URLs, command-and-control infrastructure, and known campaigns. This information can guide an investigation by showing whether an indicator has been observed by another security provider and how it was classified.

The lesson also made an important point: reputation data is supporting evidence, not final proof. A new malicious file may not yet appear in public feeds. An IP address that was malicious months ago may later be reassigned. A cached result may also be older than the activity I am investigating. I still need to check the date, source, behaviour, and internal evidence.

Threat Intel dashboard explained

Dashboard area	What I look for
Indicator search	I enter a hash, IP address, domain, or URL and make sure the indicator type is interpreted correctly.
Reputation or score	This summarises how the platform or source assesses the indicator. I treat it as one piece of evidence.
Data source	This identifies who supplied the intelligence, such as a malware-tracking feed or security vendor.
Classification and tags	Labels may describe malware, phishing, command-and-control, spam, botnet activity, or another category.
First and last seen	Dates help me judge whether the intelligence is recent and relevant to my incident.

Dashboard area	What I look for
Related indicators	Associated domains, IP addresses, hashes, or malware families can help expand the scope of the investigation.
Community or vendor observations	Comments and detection details may provide useful context, but I verify them against internal evidence.

How I use threat intelligence responsibly

- Check more than one source when the decision is important.
- Look at the observation date instead of relying only on the current score.
- Compare the external result with endpoint, network, email, and identity evidence.
- Do not declare a file safe only because no vendor detects it.
- Do not declare an IP malicious forever; ownership and usage can change.
- Record the source and lookup time in the case notes.

17.1 Threat Intelligence Lesson Question

Question

Question: What is the data source of the hash e1def6e8ab4b5bcb650037df234e2973 on the Threat Intel page?

Answer: AbuseCH

Explanation: Open the Threat Intel page, search the hash, and read the data-source field in the result.

The screenshot shows a dark-themed dashboard with a search bar at the top. The search bar contains the hash "e1def6e8ab4b5bcb650037df234e2973". Below the search bar, there are several filters: "Date range" (Set to "Select Date"), "Search by data type" (Set to "Hash"), "Search by data" (Set to "Search"), and "Search by tag" (Set to "Search"). A "Search" button is located at the bottom right of the filter section. Below the filters, there is a table with the following columns: DATE, DATA TYPE, DATA, TAG, and DATA SOURCE. The table contains one row with the following data: DATE: Oct, 11, 2020, 09:24 PM; DATA TYPE: Hash; DATA: e1def6e8ab4b5bcb650037df234e2973; TAG: malware; DATA SOURCE: AbuseCH.

DATE	DATA TYPE	DATA	TAG	DATA SOURCE
Oct, 11, 2020, 09:24 PM	Hash	e1def6e8ab4b5bcb650037df234e2973	malware	AbuseCH

Correct

What is the data source of the "e1def6e8ab4b5bcb650037df234e2973" hash on the threat intel page?

AbuseCH

Completed

Get unstuck?

Question-version note

Published notes for an older version of the lesson show a similar exercise asking for the score of the same hash rather than its data source. Reputation scores can change as feeds are updated. For that version, I would search the hash in the live dashboard and record the score shown at the time of completion. The recent exercise wording asks for the data source, which is AbuseCH.

What the exercise is really testing

The important skill is knowing where the intelligence came from and how current it is. A number without a source and date is weak evidence.

18. Common Mistakes Made by SOC Analysts

This lesson was useful because it focused on investigation habits rather than definitions. A technically skilled analyst can still reach a poor conclusion by rushing, depending on one tool, or failing to search for related activity. The mistakes below are written as practical reminders I would use during a real shift.

Common mistake	A stronger approach
Depending too heavily on VirusTotal	Use VirusTotal as enrichment, then compare the result with process behaviour, network activity, static or dynamic analysis, and internal telemetry.
Rushing sandbox analysis	Allow enough time for delayed behaviour and remember that some malware detects sandbox environments and remains inactive.
Insufficient log analysis	Search the wider environment for related IPs, domains, hashes, users, and hosts so a single-device alert does not hide a broader incident.
Ignoring the date of a VirusTotal result	Refresh the analysis when appropriate and check when the file or URL was last scanned.
Closing familiar alerts without checking	Confirm the evidence every time. A familiar alert can still represent a new or more serious event.
Writing weak case notes	Record the evidence, timeline, affected assets, final classification, and the reason for closure or escalation.
Acting before understanding business impact	Check the role of the affected system and obtain the required approval before disruptive containment actions.
Failing to share detection feedback	Report repeated false positives and explain which condition or exception would make the rule more accurate.

A personal checklist before I close an alert

- Have I checked the alert details rather than relying on the title?
- Have I searched for related activity in logs and endpoint data?
- Have I checked the age and source of threat-intelligence results?
- Have I considered a legitimate business explanation?
- Have I documented the evidence that supports my decision?
- Does the rule need tuning or a safe exception?
- Would another analyst understand my conclusion from the notes alone?

19. SOC Fundamentals Final Quiz Questions and Answers

The final quiz connects the complete course rather than testing one dashboard. I have written the answers in clear language and added a short explanation so this section reads as a learning record, not simply an answer sheet. The wording below follows a published walk through of the course; the live platform may adjust wording or answer order.

Question 1

Question: Which type of SOC team does not have its own facility and often works remotely in different locations?

Answer: Virtual SOC

Explanation: The analysts work from different locations and collaborate through shared security and communication systems.

Question 2

Question: I am responsible for connecting security products to each other. What is my job title?

Answer: Security Engineer

Explanation: Security engineers build and maintain integrations between SIEM, EDR, SOAR, log sources, and other security products.

Question 3

Question: What is SOC?

Answer: Security Operations Center

Explanation: A SOC is the function that continuously monitors, detects, investigates, and responds to cybersecurity threats.

Question 4

Question: Which tool is the most important for a SOC analyst?

Answer: All of these, and much more

Explanation: SIEM, Log Management, and EDR are all important. A strong investigation usually needs several tools and the analyst's own technical judgement.

Question 5

Question: Which SOC model consists of internal SOC personnel working with an external Managed Security Service Provider?

Answer: Co-Managed SOC

Explanation: Responsibility is shared between the organisation's own team and an external MSSP.

Question 6

Question: Which SOC position aims to find threats or weaknesses before an attacker can exploit them?

Answer: Threat Hunter

Explanation: Threat hunters proactively search for suspicious behaviour that may not have triggered an alert.

Question 7

Question: What is the goal of a SIEM?

Answer: To provide real-time event logging and support threat detection

Explanation: A SIEM centralises events and applies rules or correlations so suspicious activity can be investigated.

Question 8

Question: Which LetsDefend page represents the SIEM?

Answer: Monitoring

Explanation: The Monitoring page displays new, active, and closed alerts and supports the alert workflow.

Question 9

Question: What is an EDR?

Answer: A security solution that monitors endpoint devices and supports detection, investigation, and response

Explanation: EDR records activity on computers, servers, and other endpoints rather than monitoring only network traffic.

Question 10

Question: What can be used to conduct an analysis with SOAR?

Answer: Playbook

Explanation: A playbook defines the checks and response steps that the platform and analyst follow.

Question 11

Question: Which information is not normally provided by a Threat Intelligence Feed?

Answer: A sample of the infected file

Explanation: Feeds usually provide indicators, reputation, classifications, and context, but they may not provide the original infected sample.

Question 12

Question: Which is a common mistake for SOC analysts?

Answer: Insufficient log analysis

Explanation: Failing to search related logs can hide other affected devices and lead to an incomplete incident scope.

Question 13

Question: What are the traditional NIST incident-response lifecycle phases used in this course?

Answer: Preparation; Detection and Analysis; Containment, Eradication and Recovery; Post-Incident Activity

Explanation: The quiz uses the traditional four-phase incident-response model taught in the course.

How I would use this section

I would review the explanation column before retaking the quiz. Memorising an answer may help once, but understanding why it is correct makes the knowledge useful in a real SOC.

20. Overall Learning Outcomes

What I can now explain

- How a Security Operations Center is organised and how the main SOC roles work together.
- How a SIEM rule turns events into an alert and why the alert is only the start of an investigation.
- How to search central logs using fields, values, time ranges, and raw events.
- How to investigate endpoint processes, command lines, files, network connections, and scope with EDR.
- How SOAR playbooks connect tools and standardise repeatable investigation tasks.
- How to use threat intelligence as supporting evidence without treating a score as absolute proof.
- How common investigation mistakes lead to false confidence, incomplete scope, or poor documentation.

- How to document, close, escalate, and improve an alert investigation.

Practical skills I still want to develop

- Writing more advanced log queries and correlating records from several data sources.
- Reading Windows process trees and recognising common living-off-the-land techniques.
- Using EDR live-response commands safely in a controlled lab.
- Building or editing a SOAR playbook and understanding approval gates.
- Comparing multiple threat-intelligence sources and recording confidence clearly.
- Writing concise case notes that are useful to a senior analyst or incident responder.

Source and accuracy note

The official LetsDefend SOC Fundamentals course page confirms that the course includes SIEM, Log Management, EDR, SOAR, Threat Intelligence Feed, common analyst mistakes, and a final quiz. The lesson explanations in this portfolio were rewritten in my own words from the course material supplied for this project. The practical question values were cross-checked against LetsDefend community discussions and published course walkthroughs. Because training datasets and interfaces can change, I would verify any dynamic value against my own logged-in dashboard before final submission.

Official course page: <https://app.letsdefend.io/training/lessons/soc-fundamentals>

SIEM lesson page: https://app.letsdefend.io/training/lesson_detail/siem-and-analyst-relationship

LetsDefend community FAQ: <https://forum.letsdefend.io/t/faq-soc-fundamentals/128>

Log Management FAQ: <https://forum.letsdefend.io/t/faq-log-management/31>

EDR FAQ: <https://forum.letsdefend.io/t/faq-edr-endpoint-detection-and-response/32>

Final reflection

The complete course showed me that SOC work is not about clicking one tool until it gives an answer. It is about combining evidence from several systems, checking the context, documenting the reasoning, and knowing when to ask for help or escalate.